Universität Freiburg – Institut für Physik

Applied Theoretical Physics

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Computational Physics: Materials Science - Exercise 6, SS 2026

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In this exercise, we will take a look at the diffusion of particles in computer simulations. We will make use of the Einstein equation and the Green-Kubo relation to calculate the self diffusion coefficient of our system.

Enrich your existing MD simulation by implementing the Lennard-Jones (12-6) pair-potential (LJ potential), given by equation (1)

$$V_{\mathrm{LJ}}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^{6} \right], \tag{1}$$

where ε is the potential well depth, σ is the particle size parameter and $r = |\mathbf{r}_i - \mathbf{r}_j|$ is the pair-particle distance. Include the analogous force calculation to determine the force acting on each particle, each timestep. Respect the minimum-image convention and equilibrate your simulation before collecting data. Make sure to frequently save the position and velocity of your particles. You can use the parameters provided in table 1 for a successful simulation. Monitor energy and other quantities of interest to judge if the amount of equilibrium steps n_{eq} is sufficient for your simulation setup.

Problem 6. Diffusion coefficient in liquid molecular dynamics simulations

Our simulations should be efficient at this point (make use of **numba**), so use N=125 particles at least and equilibrate your simulation for n_{eq} steps before starting your production run with n_{prod} steps. You can complete this exercise with either thermostat you have implemented in the previous exercise sheet. Make sure to understand the implications of each thermostat on the diffusion coefficient.

(a) After your run is finished, calculate separately the diffusion coefficient via the Einstein equation using the MSD, see equation (2). Let us first make a mistake and use *wrapped* coordinates. Show an evocative plot and explain why this approach is incorrect.

$$\mathrm{MSD}(t) = \left\langle |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle_i \tag{2}$$

How can you modify this equation to make the MSD statically more stable? How can you speed up your MSD calculation drastically? Plot the MSD in a log-log plot and indicate with two characteristic slopes what regimes you observe. This plot serves as a validation that your simulation works as expected.

(b) Calculate the diffusion coefficient from the MSD using the Einstein equation. Report the diffusion coefficient when you change the equilibration/production time. What can you conclude?

(c) Calculate the diffusion coefficient using the Green-Kubo relation, which is the time-integral over the velocity autocorrlation function:

$$D = \frac{1}{3} \int_0^\infty \langle \mathbf{v}_i(0) \cdot \mathbf{v}_i(t) \rangle_i \; dt, \tag{3}$$

where $\mathbf{v}$ is the velocity of a particle. How does it compare to your previously calculated diffusion coefficient. Why is the diffusion coefficient obtained via equation (3) difficult to converge properly?

In the following we will focus on the diffusion coefficient obtained via the Einstein equation. We will analyse how the diffusion coefficient changes when varying different parameters of our simulation. Make sure to vary each parameter in a meaningful way, so that your parameter sweep exhibits a clear scaling law:

(d) Repeat your simulations for different densities ρ. How does the diffusion coefficient change?

(e) Analyse how your diffusion coefficient changes if you increase the size of the simulation box, while keeping the density ρ fixed. Verify a known scaling law for the size dependent diffusion coefficient. How could you extrapolate the real diffusion coefficient from your size-dependent diffusion coefficient?

(f) How does the diffusion coefficient depend on temperature? Prove that your diffusion coefficient follows the Arrhenius equation.

What to submit

Hand in a zip file on ILIAS containing the following files:

- your Python source code
- a short report in PDF format with your main results

A good submission contains:

- a brief summary of the code base so far with an explanation on what you added to your simulation.
- plots which show your results in a clever way and demonstrate the expected scaling laws. If points do not follow theory, make reasonable effort to explain why.
- a snapshop from ovito to show that your simulation works.

How to prepare for the exercise group

- Make sure you can explain your results and the code.
- Keep your chat history with an AI chatbot to explain in the tutorial how you **interacted** with it to develop your code.
- Prepare yourself for the questions outlined below.

Questions to think about

Make sure to feel comfortable talking about your code, your results and the questions below. These questions provide the key take-away points of this exercise:

- What is the Lennard-Jones potential and where is it used? What is the main difference between the Morse potential and the LJ potential in terms of atomistic simulations?
- Explain different approaches to calculate the diffusion coefficient in a system.
- Establish a link between microscopic interactions and macroscopic measurable quantities. What other physical quantities can you derive with these relations?

Parameter	**Value**
N	125
n_{eq}	10000 steps
n_{prod}	10000 steps
m	39.95 g/mol
T	300 K
ε	$0.3\, k_B T$ (kcal/mol)
σ	0.341 nm
Δt	2 fs
ρ	0.5 σ^{-3}

Table 1: Simulation parameters for this exercise sheet.